Concept of a function → 06.10.2025

Concept of a fn is one of the most basic in all math.

- The meaning of the word "function" has evolved & "changed during the last three centuries."

Modern Meaning → is a much broader & deeper than its elementary meaning from earlier days!

the Statement → "y is a function of x" means

something very much like "y is related to x by some formula

↳ Some idea about a function → but it is incomplete.

In traditional algebra → x & y stand for numbers.

But → today → functions can be defined that have nothing to do with numbers →

In the study of Calc 06.10.2025

you shall be mostly concerned with functions, which are related to numbers

like any other concept in Math →

"The concept of function is nicely expressed through the language of sets."

Look up Set Theory

1) ordered pairs ii) Cartesian product sets

These terms are needed to define a "function" on the basis of set theory

i) Ordered Pairs: When considering a pair of things as a whole

↳ use the terms couple or just pairs

If $A = \{1, 2, 3, 4\}$ then the subsets $\{1, 2\}$, $\{1, 3\}$ $\{1, 4\}$

$\{2, 1\}$ 06.10.2025

↳ $\{3, 1\}$ are some examples of pairs

Here → you have listed some <u>pairs twice</u>

e.g. ↳ $\{1, 2\} = \{2, 1\}$ ;

$\{1, 3\} = \{3, 1\}$

Here the order, in wich the elements of a set are written, is immaterial.

If in a pair you wish to single out one element as being the first, then the other element becomes the second.

Once you define the procedure of fixing the position of first element (in a pair) you have an example of an ordered pair.

To denote the ordered pair → use the following notation: ↓

The ordered pair consisting of the element 06.10.2025

1, 2 → in which 1 is the first element will be written as (1, 2), whereas the ordered pair consisting of the elements 1 & 2 in which 2 is the first element → will be written as (2, 1)

so then $(1, 2) \neq (2, 1)$

∴ Def: Cartesian Product of two Sets A & B: Let A & B be two sets, You define the Cartesian product of A & B, written $A \times B$, to be the set of ALL ordered pairs $(x, y)$ where $x \in A$ & $y \in B$

Thus, $A \times B = \{(x, y) \mid x \in A \text{ \& } y \in B\}$

e.g. ⇝ let $A = \{1, 2, 3\}$ & $B = \{5, 6\}$

Then ⇝ $A \times B = \{(1, 5), (2, 5), (3, 5),$

$(1, 6), (2, 6), (3, 6)\}$

Just Remember that the product $A \times B \rightarrow$ the first element in each ordered pair belongs to $A$ & the second element to set $B$. ★ Also note that if set $A$ contains $m$ elements & set $B$ contains $n$ elements, $\rightarrow$ then set $A \times B$ will have $m \cdot n$ ordered pairs.

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## 2.2. Equality of Ordered Pairs

Two ordered pairs $(a, b)$ & $(c, d)$ are equal if $a = c$ & $b = d$

## 2.3. Relations & Functions

You know that if set $A$ contains $m$ elements & set $B$ contains $n$ elements, then $A \times B$ will have $m \cdot n$ ordered pairs. Any subset of these ordered pairs is called a relation from $A$ to $B$

Consider $\Rightarrow$ Let $A = \{1, 2, 3, 4\}$ & $B = \{2, 4, 5\}$

Then, $A \times B$ 06.10.25

$= \{(1,2), (1,4), (1,5), \ldots (4,2), (4,4), (4,5)\}$

There are many relations from $A$ to $B$

$R_1 = \{(1,2), (1,5), (2,2), (3,4), (3,5)(4,5)\}$

$R_2 = \{(1,4), (4,2), (4,5)\}$

$R_3 = \{(3,2), (3,4), (3,5), (1,4)\}$

$R_4 = \{(1,4), (2,5), (3,2), (4,4)\}$

$R_5 = \{(1,2), (2,5), (3,4), (4,4)\}$

However → if you select the ordered pairs in such a way that:

i) their first elements constitute the entire set $A$, &

ii) no two distinct pairs have the same first element.

then such a collection of ordered pairs ( from the set $A \times B$ ) constitute → a special relation from $A$ to $B$ which is called a function from $A$ to $B$

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## 2.3.1 > Domain of a Relation

In any relation ( in the form of a set ordered " " pairs ) the set consisting of the "first" element of each pair constitutes the domain of the relation.

Consider the following (above) relations

- The domain of $R_1 = \{1, 2, 3, 4\} = A$

!! But there are two ordered pairs (1, 2) (1, 5) in which the first element is the same → Hence, $R_1$ Does NOT represent a function!

- The domain of $R_2 = \{1, 4\}$ $\neq A$. 06.10.25

Also, there are two ordered pairs $(4, 2)$ & $(4, 5)$ in which the first element is the same. Therefore, $R_2$ is NOT a function from $A$ to $B$, & $R_3$ is not a function.

- The domain of $R_4 = \{1, 2, 3, 4\} = A$.

And no two distinct pairs have the same first Element. Hence $R_4$ represents a function

& $R_5$ also represents a function.

You can still define many functions from $A$ to $B$ → like so

$f_1 = \{(1, 2), (2, 2), (3, 2), (4, 2)\}$,

$f_2 = \{(1, 4), (2, 2), (3, 5), (4, 4)\}$,

$f_3 = \{1, 5), (2, 4), (4, 2), (3, 2)\}$,

↓

$f_4 = \{(1, 5), (2, 5), (4, 5), (3, 5)\}$ 06.10.2025

& so on

OK, now let's define a "function" on the basis of set theory!

(2.4) Def.

Let $A$ & $B$ two nonempty sets.

A function $f$ from $A$ to $B$ is a subset of $A \times B$ (involving the entire set $A$) with the property that each "a" belonging to $A$, belongs to precisely one ordered pair $(a, b)$, in the subset of $A \times B$, under consideration.

>> meaning, a function $f$ from set $A$ to set $B$ consists of a set of ordered pairs $(a, b) \in A \times B$ such that no two ordered pairs have the same first element!

## Alternative Definition of a "Function"

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2.4.1

A function $f$ from set $A$ to set $B$ (written → $A \to B$) is a rule of correspondence that associates to each element of $A$, one and only one element of $B$

(A function is called a mapping from $A$ to $B$)

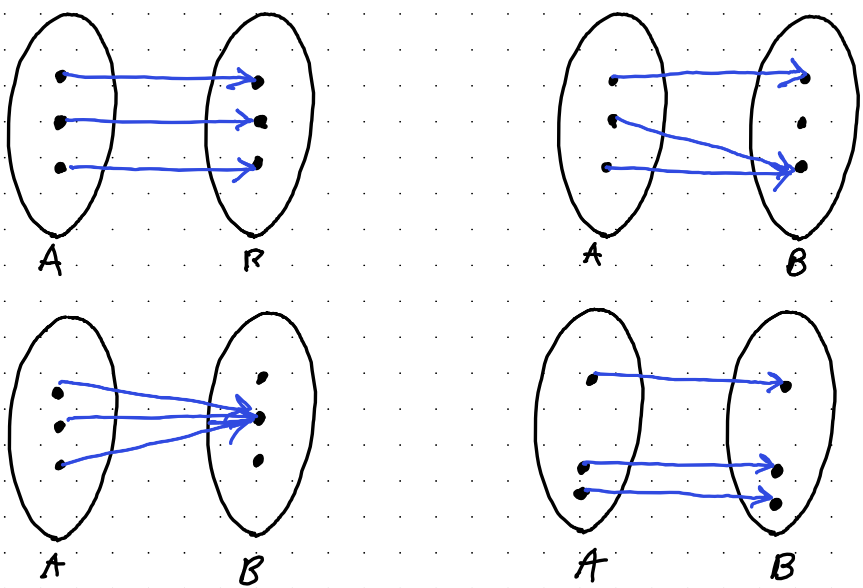
Observe that →

i) Each Element of $B$ need not be in association, but every element of $A$ must be involved in it. Hence, a function is a one way pairing process.

Every element of $A$ pairs off with some element of $B$ → but not in converse.

ii One element of A cannot be associ-ated to more than one element of B, but one element of B may correspond to two or more elements of A. 06.10.2025

The correspondence from the elements of set A to set B → represents function(s)



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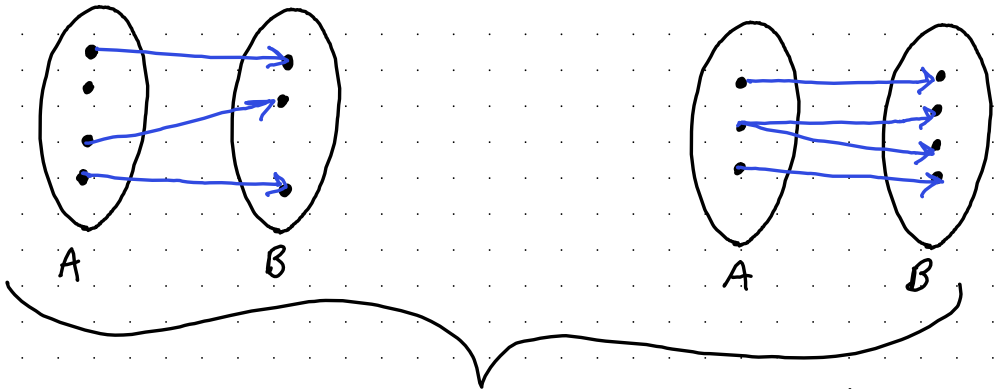


## 2.5 Domain, Codomain, Image, & Range of A function

let $f$ be a function from set $A$ to set $B$ ($f: A \to B$), then

- The <u>entire</u> set $A$ is called the domain of $f$
- The <u>entire</u> set $B$ is called the codomain of $f$
- An <u>element</u> of $B$ that corresponds to some element $x$ of $A$ is denoted by $f(x)$,

& its called the image of $x$ under $f$ 06.10.2025

- The set of all images constitute the range of $f$. The range of $f$ is denoted by $f(A)$ & it is a subset of set $B$.

In other words → $f(A) \subseteq B$

## 2.6 Distinction Between "$f$" And "$f(x)$"

You write $y = f(x)$ → because a function $f: A \to B$, let $(x, y)$ be an <u>arbitrary</u> ordered pair belonging to $f$.

Then instead of writing $(x, y) \in f$, write as

$y = f(x)$ → meaning $y$ is related to $x$, through "$f$"

This is read as $y$ [or $f(x)$] is a function of $x$.

<u>$x$ represents an arbitrary element</u>

of the domain & y represents the corresponding element of the range.



Remember that the element "$f(x)$" is selected by the rule of correspondence defined by the function "$f$." Thus, "$f$" represents the rule of correspondence.

The symbol $f(x)$ represents the result of applying "$f$" to "$x$." You call it the value of "$f$" at $x$.

If "$f$" is the function defined by the formula $x^2 - 2x + 3$, then you usually write

$$y = f(x) = x^2 - 2x + 3, \quad (x \in A)$$

this tells you that the rule of correspondence is the formula, "$x^2 - 2x + 3$" → which converts every number $x \in A$, into a new number "$f(x)$" belonging to $B$

↑ using the formula

$\phi 6.1\phi$
$2\phi 25$

Thus → the formula "$x^2 - 2x + 3$" must be identified with the function "f." } for computing the value "$f(x)$" → use the formula defining "f"

→ you choose $x = \phi \in A$, you get the corresponding element $y \in B$ as $f(\phi) = 3$ you will also get for $x = 1$, you will get $y = 2$, for $x = 5$, then $y = 18$

To avoid the confusion → possibly caused by (due to equation → "$x^2 - 2x + 3$") wherein $f(x)$ is equated to $x^2 - 2x + 3$ you should write it as →

$x \to x^2 - 2x + 3$, which states that "f" is a function that converts each $x$ into $x^2 - 2x + 3$

↓

# Dependent & Independent Variables 06.10.20

2.7 When the rule for a function is given by an equation of the form 2.5

$$y = f(x)$$

e.g ⇒ $y = x^2 - 2x + 3$ or $y = \sin x$ or $y = e^x$

then $x$ is called the independent variable & $y$ [ or $f(x)$ ] → is called the dependent variable.

For the dependent variable $y$ [or $f(x)$] you look at the expression for $f(x)$

## 2.8 Functions at a Glance

(1) A function consists of three things :

i) A set Known as the domain of the function.

ii) A set Known as the range of the function.

iii) A correspondance ( A rule or a method or a procedure ), which associates with each member of the domain, precisely one member of the range.

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2). If you conceive of a function as being specified by a set of ordered pairs, then you must insist that no two selected distinct pairs may have the same first element.

## (2.9) Modes of Expressing A Function

A function is completely known if the objects & corresponding images are known.

There are many ways in which this can be done!

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# four Methods for describing a function

## i) Statement of the Rule of Association (By Formula or Otherwise):

If the domain of the function is Known & rule(s) of association between objects & images are Known, the images can be found out & the correspondence is completely Known.

As you go through the other modes of expressing a function, it will be realized that this mode of expressing a function is the most accurate & complete

{{ when the domain is an infinite set }}

## ii Description by Tables

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When the domain of a function consists of small number of elements (e.g → 10 - 20 numbers), it may be described by tabulating objects & their corresponding images!

>> Which is very useful when the objects & Images cannot be connected by a fixed rule due to irregular variations, & so on (This mode of expressing a function however, is not useful when the domain has a large number of elements)

## iii) Description in Terms of Ordered Pairs

: A function is expressible as a set of ordered pairs. In fact, the set-theoretic definition of a function is the basis for this mode of expressing a function.

↑ not that useful in Calculus

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$\boxed{\text{iv) Description by Graphs}}$

you know that a function is a set of ordered pairs & each ordered pair is liable to be represented as a point in the plane.

Therefore, the function itself is represented by the set of these points.

If $f: A \to B$ is a function then the the set

$$\{(a, f(a)) \mid a \in A\}$$ is called

the graph of $f$ & is a subset of $A \times B$

↓

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In particular if $A, B \subseteq \mathbb{R}$,
The graph of $f$ can be represented by points in the plane.

The graph of a function may be a set of distinct points or it may be a (continuous) curve.

Consider the fn $\rightarrow$ $f(x) = 2x + 1$, $x \in \{0, 1, 2, 3\}$

Here $\rightarrow$ $A = \{0, 1, 2, 3\}$

The graph of this function consists of 4 isolated points

$(0,1), (1,3), (2,5), (3,7)$

& is not a continuous curve.

However! the graph of the function

$$f(x) = 2x + 1, \quad x \in \mathbb{R}$$

is a contin curve (line) passing through above four points!

2.1Ø Types of Functions Ø6.1Ø.2Ø25

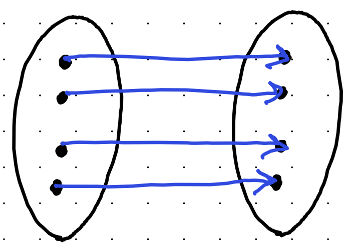
You know that a relation $f: A \to B$, which satifies the following two conditions

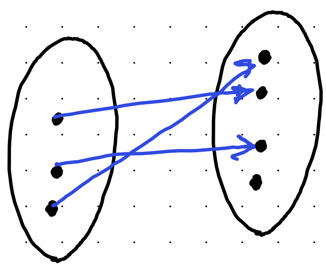
↳ will be called a function

I) Each element of the domain $A$ is involved in the relation.

II) Each element of $A$ is associated to exactly one element of $B$ & not more than one element of $B$

Injective functions





Note : That both the specification 06.10.20 25 (S) are imposed on the elements of A, & that no restriction is imposed on the elements of the codomain B.

B) If you make restrictions (I) & (II) on the codomain B, you get two special kinds of functions namely (i) one - one function & (ii) onto function 2>

A) One - One function : A function is one - one provided distinct elements of the domain are related to distinct element of the codomain, → a function $f: A \to B$

defined to be one-one if the images of distinct element of $A$ under $f$ are distinct, that is for every $a_1, a_2 \in A$, 06.10.2025

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$$

★ which also means that $f(a_1) \neq f(a_2) \Rightarrow a_1 \neq a_2$

↳ A one-one function is called $\boxed{\text{injective function}}$

★ If there is at least one pair of distinct elements,

$a_1, a_2 \in A$ such that →

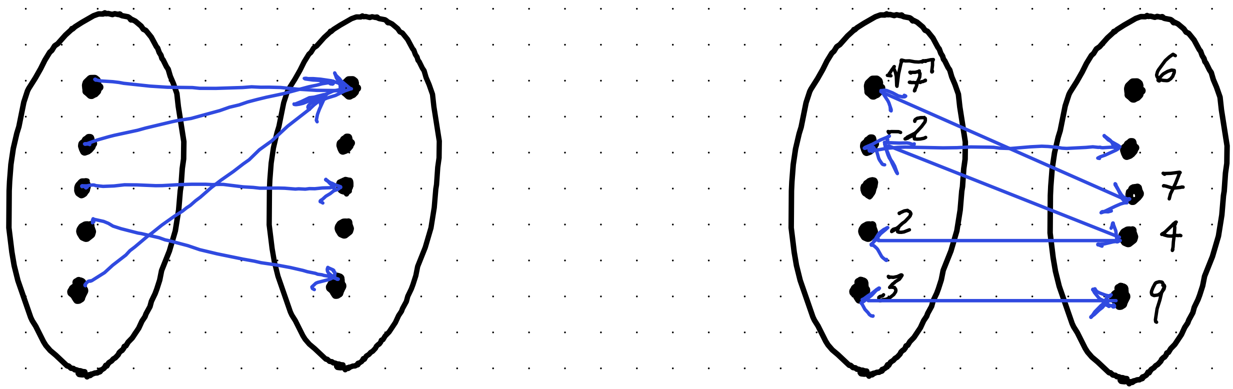
$$f(a_1) = f(a_2) \text{ [ through } a_1 \neq a_2 \text{]}$$

then such a fn is called <u>many-one</u>!

you define many - one function as follows: 06.10 2025

## B) many - One Functions:

If the codomain of the function has at least one element, which is the image for two or more elements of the domain then the function is said to be many - one function.




Many to ONE functions

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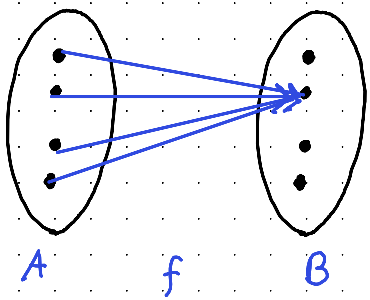


Constant function

(many - one function

A constant function is a special case of many - one function

Ⓒ Onto Function : A function $A \to B$ is called an onto function if each element of the codomain is involved in the relation

(Here, range of $f$ = codomain $B$)

In other words → a function $f: A \to B$ is said to be onto if every element of $B$ is the image of some element of $\underline{\underline{A}}$

, under $A$, under $f$, that is for every $b \in B$, there exist an element $a \in A$, such that $f(a) = b$

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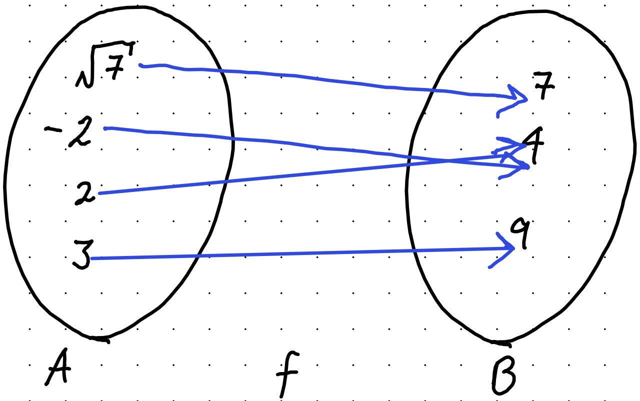
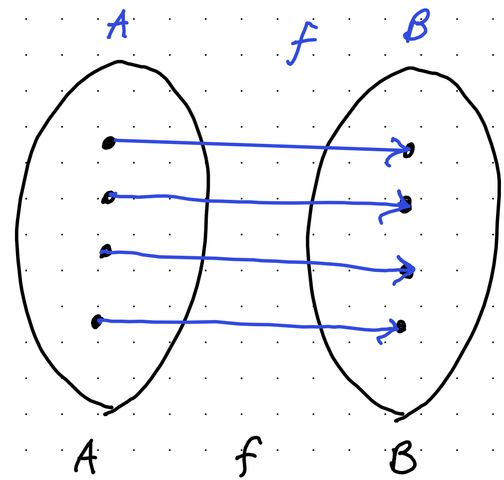



$f: x \to x^2$

(a)




## Surjective functions

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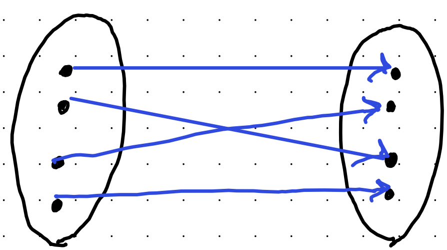
Onto function is called a surjective function.

"The most important functions are those which are both one-one & surjective
(and one to many)

onto

Remember →, In a function that is one-one & onto, each image corresponds to exactly one element of the domain & each element of codomain is involved in the relation. Such a function is also called one-to-one correspondence

☆ Bijective function



One to One, & an onto function (surjective function)

Def: Consider a function $f$: $A \to B$ with "A" as the domain of the definition (i.e. the admissible set of the values of $x$ & "B" as the range (i.e. the corresponding values of $y$) 06.10.20 25

you say that the function $y = f(x)$ specifies a mapping of the set "A" onto the set B: if two different points of the set A, there corresponding different points of the set B, & the entire set B: is the rangie of "f."

Such a mapping is called one-to-one mapping

Def 2: let A & B be two nonempty sets. A rule f, which, associates with each element "a" of the set A, exactly one element "b" of the set B,

& under which, each element "b" of set B, of the set B, under which, each element "b" of set B corresponds to exactly one element "a" of the set A, is called a one-to-one correspondence between the sets A & B

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e.g. → Consider the fn $y = f(x) = x^3$

"Hence, for every value of $x \in \mathbb{R}$, there corresponds a single value of $y$, & conversely, to each $y \in \mathbb{R}$, there corresponds a single value of $x$ given by $\sqrt[3]{y}$. Therefore, $f$ specifies a one-to-one mapping, from $\mathbb{R}$ onto $\mathbb{R}$

e.g → Consider the fn $y = g(x) = x^2$. Here, for every value of $x \in \mathbb{R}$, there corresponds a single value of $y \in (0, \infty)$. However, to every $y > 0$, there correspond two values of $x$: $x = \pm\sqrt{y}$ Therefore, "g" <u>is not</u> a one-to-one correspondence.

Consider the exponential function $y = f(x) = e^x$. It can be shown

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that the function $f(x) = e^x$ is one-to-one mapping from $(-\infty, \infty)$ onto $(0, \infty)$. Note → that for $x_1 \neq x_2$, you have $e^{x_1} \neq e^{x_2}$ where $x_1, x_2 \in \mathbb{R}$ & $e^{x_1}, e^{x_2} \in \mathbb{R}^+$

Consider → $\dfrac{e^{x_1}}{e^{x_2}} \neq 1 \Rightarrow e^{x_1 - x_2} \neq 1$ or $e^{x_1 - x_2} \neq e^0$

since $e^0 = 1 \Rightarrow x_1 - x_2$ $x_2 \neq 0 \Rightarrow x_1 \neq x_2$

In other words → $e^{x_1} \neq e^{x_2} \Rightarrow x_1 \neq x_2$

Thus, $x_1 \neq x_2 \Leftrightarrow e^{x_1} \neq e^{x_2}$

Therefore, "f" defines a one-to-one correspondence from $(-\infty, \infty)$ onto $(0, \infty)$ ( Here it is important to note that, a one-to-one mapping has defined from the entire real line on the positive part of the real line.

# II Inverse Function $f^{-1}$ 06.10.25

If a function "$f$" is one-to-one & onto, then the correspondence associating the same pairs of elements in the reverse order is also a function. → fn denoted as $f^{-1}$ → called the inverse of the function $f$. $f^{-1}$ is also one-to-one & onto.

Remark → A function $f$ has an inverse provided that there exists a function, $f^{-1}$ such that

i) the domain of $f^{-1}$ is the range of $f$
& ii) $f(x) = y$ if & only if $f^{-1}(y) = x$ for all $x$ in the domain of "$f$" & for all $y$ in the range of "$f$"

Remember → Not every fn has an inverse. If a function $f: A \to B$ has an inverse, then $f^{-1}: B \to A$ is defined, such that, "the domain of $f^{-1}$ is the range of $f$, & the range of $f^{-1}$ is the domain of $f$", associating the same pairs of elements.

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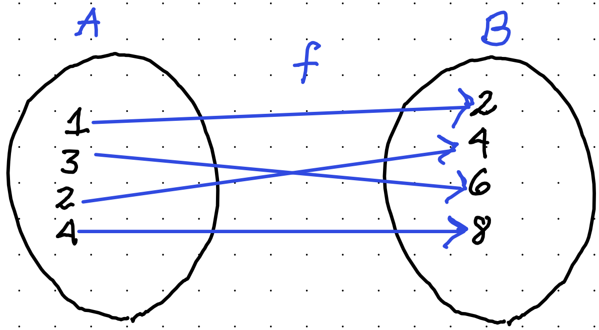
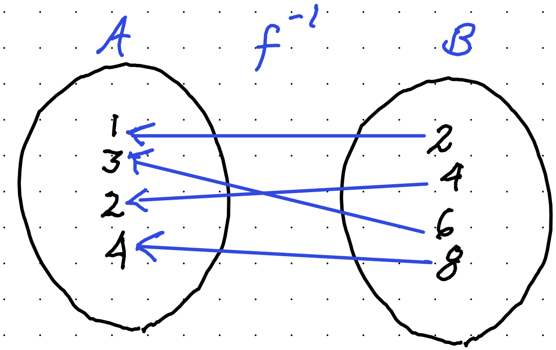



$f: A \to B$

one-one & onto




$f^{-1}: B \to A$

one-one & onto

It can be shown that if has an inverse, then the inverse function is uniquely determined

Sometimes → you can give a formula for $f^{-1}$

e.g → $y = f(x) = 2x$ then $x = f^{-1}(y)$
$= (1/2)y$

Similarly, if $y = f(x) = x^3 - 1$ 06.10.2025

then $x = f^{-1}(y) = \sqrt[3]{y+1}$, in each case,

you simply solve the equation that determines $x$ in terms of $y$.

The formula always give the formula for $f^{-1}$

e.g. → consider the function

$y = f(x) = x^5 + 2x + 1$ It is beyond your capabilities to solve this equation for $x$

("and why?")

In such cases, you cannot decide whether a given function has an inverse or not.

There are criteria that tell whether a given function $y = f(x)$ has an inverse, irrespective of whether you can solve it for $x$.

In the case of simple functions

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↳ such as linear functions etc(?)
there is a 3-step process that gives a formula for the inverse.

1) Solve the equation $y = f(x)$ for $x$, in terms of $y$

2) Use the symbol $f^{-1}(y)$ to name the resulting expression in $y$.

3) Replace $y$ by $x$ to get the formula for $f^{-1}(x)$

A general polynomial equation of degree $\geq 5$ cannot be solved in terms of the coefficients

"$f$" be strictly monotonic (either strictly increasing or strictly decreasing e.g. Consider the fn $06.10.2025$

$$y = f(x) = 3x - 2, \quad x \in \mathbb{R}$$

let's find the inverse

↓ Sol → step 01

① $y = f(x) = 3x - 2 \rightarrow 3x - 2 = y + 2$

$+2$

② $f^{-1}(y) = \frac{y+2}{3}$ $\qquad \frac{3x}{3} = \frac{y+2}{3} = x = \frac{y+2}{3}$

③ $f^{-1}(x) = \frac{x+2}{3}$

e.g (2) Find the formula for $f^{-1}(x)$ if

$$y = f(x) = \frac{x}{1-x}$$

↓

$$x = y(1-x)$$

$$\frac{x}{1-x} = y =$$

$$x = y(1-x), \quad (y - yx) = x$$

$$x = y - yx$$

$$x = y(1-x)$$

X

Observe : → you see that in this expression ( of the inverse, function) the roles of vars $x$ & $y$ are interchang-ed

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**Both** the functions describe one & the same curve in the $xy$ - plane

→ they are said to be mutually inverse functions.

For the function $f$, the axis of the independ-ent variable is the $x$ - axis, while for the function $f^{-1}$ "this role" is played by the $y$-axis.

If you want to construct the graphs mutually inverse functions → so that the axis of args → i.e the axis of independent variables) for both of them is the $x$-axis then denote the independent variable in

$x = f^{-1}(y)$ by $x$ & express the inverse function in the form

↳ $y = f^{-1}(x)$

** In this notation, $x = f^{-1}(y$ $\qquad$ 06.10.25

↳ $y = f^{-1}(x)$ ↓ → the letter $x$ designates the independent variable & the letter $y$ the dependent variable; for both the mutually inverse functions.

Thus the functions → $y = x^3$ &

$y = \sqrt[3]{x}$ → represent a pair of mutually inverse functions

↳ also → $y = 10^x$ & $y = \log_{10} x$

are mutually inverse functions

There is a simple relationship between the graphs of two mutually inverse functions

$y = f(x)$ & $y = f^{-1}(x)$

They are symmetric with respect to the line $y = x$

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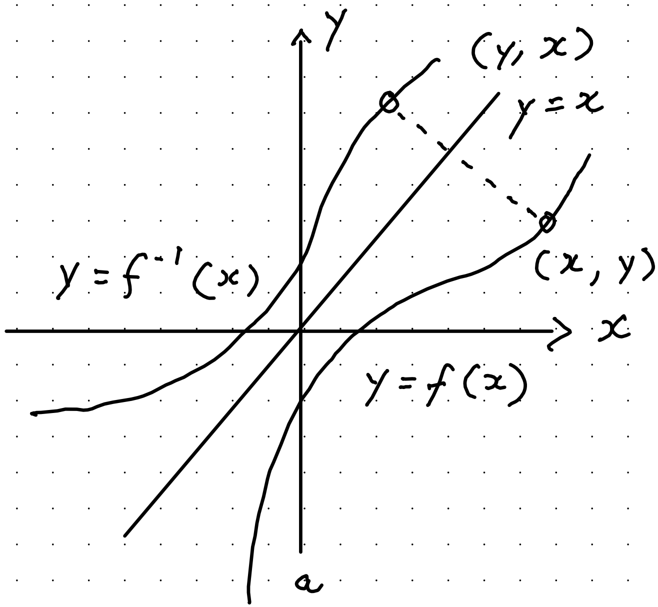
you can interchange the roles of $x$ & $y$ on a graph → to reflect the graph across the line $y = x$

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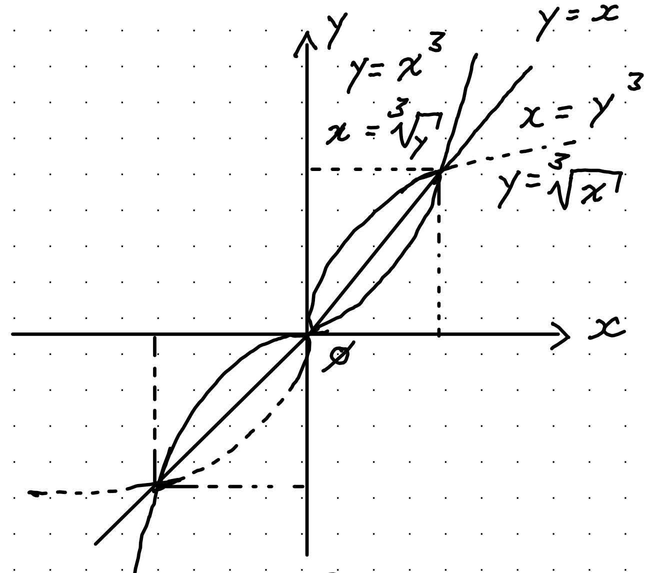


B

Reflected across the line $x$

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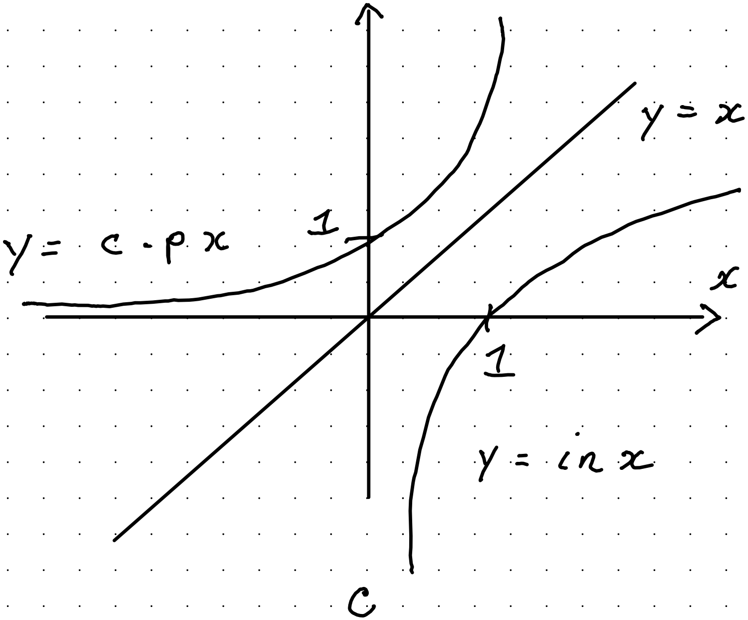


Reflected across the line x

don't confuse $f^{-1}$ with $\frac{1}{f}$

Remember → Each pair of inverse functions i.e. $f$ & $f^{-1}$ behave in such a way that one function undoes (or reverses) what the other does → $f(x) = y$, →

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then $f^{-1}(f(x)) = x$ & if

$f^{-1}(y) = x$ $\Rightarrow$ then $f(f^{-1}(y)) = y$